Silicon & Handset — Post-v0.3.0 Technical Audit, Architectural Remedies & Implementation Roadmap

Document Purpose: AI-to-AI technical handoff, post-v0.3.0 weakness audit, causal system design, and phased implementation specification for coding agents (`Codex`, `Agy`, `Gemini`).
Target Branch / Source Root: `spinoff-phone` (`SPINOFF_SOURCE/`)
Assumed Baseline: `v0.3.0` (`ACTIVE_PROMPT.md` scope assumed complete and passing all acceptance checks).
Governing Rules: `AGENTS.md` (immutable), `BRANCH_LAYOUT.md`, `.agents/skills/silicon-handset-dev/SKILL.md`, `.agents/skills/silicon-handset-every-patch/SKILL.md`.

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0. Baseline Context & Non-Negotiable Invariants

0.1 Assumed v0.3.0 Baseline State

This specification assumes the `v0.3.0` candidate assignment in `ACTIVE_PROMPT.md` has been fully implemented and verified:

1. **Explicit Component SKU & Lot Provenance:** Generic `kind:generation` fungibility has been replaced with explicit component SKUs, design ownership, IP/licensing rights, physical production lots, revisions, and qualification states.
2. **Non-Retroactive Physical Stock & BOM Binding:** In-flight shipments, reserved BOMs, active WIP batches, and assembled handsets retain exact historical SKU/lot identities when newer revisions are researched.
3. **Divergent Firm-Specific R&D Portfolios:** Companies pursue distinct, resource-constrained engineering targets (compute, efficiency, yield, optics/ISP, thermal envelope, memory/storage, battery/BMS, OS/BSP) driven by named staff subjects, potentials, lab tooling, and NRE budgets.
4. **Autonomous Make / Buy / License / Contract Execution:** Seeded rival brands (4–8) and independent suppliers operate without player-relative rubber-banding or artificial catch-up.
5. **Trustworthy Thai Observer UI & Edit Hub:** All player-facing labels, causal explanations, SKU provenance views, and logged `Edit Hub` experiments render in clear Thai without exposing operational management buttons.

0.2 Immutable System Invariants (Must Hold in Every Future Patch)

- **Locked Human Core:** Never modify `AGENTS.md`, `SPINOFF_SOURCE/js/human-core.js` (SHA-256 `fa83eaa0ef3407f26f21675161f632a4365242f67fea7b0c192e59e75b9d5c32`), or `SPINOFF_SOURCE/CORE_LOCK.json`. Always refreeze profiles via `H.freezeProfile()` on deserialization and permitted `Edit Hub` trait edits.
- **Observer-Only Gameplay:** Never add manual corporate control buttons (approve project, buy stock, set price, hire staff, trigger launch). NPCs execute all operational and strategic decisions autonomously; the user controls time, inspection views, and logged `Edit Hub` condition experiments.
- **Consolidated Satang Conservation:** Every monetary value must be a non-negative safe integer in satang. `totalCash(w) === w.openingCapital + w.editCapital` must hold after every tick, transaction, refund, escrow settlement, failure, and cancellation. Zero free cash refills or hidden subsidies.
- **Physical Material & Lot Conservation:** Raw inputs, wafer/component lots, transit shipments, reserved BOMs, WIP, scrap, active installed base, RMA replacements, and finished goods must reconcile identically in `audit(w)`.
- **Partition-Invariant Determinism & Read Purity:** `advance(w, 60)` must produce byte-identical state to twelve calls of `advance(w, 5)`. Read-only queries (`snapshot(w)`, `consumerUtility()`, quote/feasibility inspectors, UI rendering) must never consume simulation RNG or mutate simulation state.

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1. Post-v0.3.0 Structural Weakness Diagnosis

Inspecting `SPINOFF_SOURCE/js/sim.js`, `SPINOFF_SOURCE/js/phone-engineering.js`, and `SPINOFF_SOURCE/js/ui.js` against the long-term vision in `AGENTS.md` and `CODEX_REVIEW_AND_DESIGN_REFINEMENTS.md` reveals **six structural bottlenecks** that remain even after `v0.3.0` is completed:

Weakness W1 — Macroeconomic Cash-Sink Starvation (Zero Circular Money Velocity)

- **Root Cause in Code:** Consumer cohorts (`w.segments`: `value`, `photo`, `performance`) are initialized with static finite wallets (`500M + 400M + 300M = 1,200,000,000` satang) with invariant `s.cash + s.spent === s.initialCash`. Meanwhile, daily wages accumulate in `people[].cash` (minus `2,000` satang meals paid to `w.treasury`), and firms pay `w.treasury` for upstream raw inputs, NRE/tooling, and redesigns.

- **Failure Mode:** `people[].cash` and `w.treasury.cash` act as monotonic liquidity sinks. Over multi-year simulations (Year 2–5), household cohort wallets drain toward zero, collapsing industry-wide handset sales regardless of product quality or price reductions.

Weakness W2 — Terminal Point-of-Sale Disappearance (No Installed Base, Aging, RMA, or OTA)

- **Root Cause in Code:** In `settleMarket()`, sold units decrement `d.inventory` and increment `d.sold`, then cease to exist as simulated entities.
- **Failure Mode:**
  - **No Endogenous Replacement Cycle:** Cohorts have no active device holdings (`installedBase`) that degrade thermally, chemically (`battery.cycleLife`), or software-wise (`os.supportMonths`) to drive realistic upgrade timing.
  - **Zero Cost of Poor Reliability:** Shipping a thermally throttled (`surfaceC > 45°C`) or low-protection handset incurs zero post-sale warranty/RMA cost, field failure rate, or brand-trust penalty.
  - **No Post-Launch Software Lifecycle:** Software engineers cannot ship OTA firmware/BSP updates to reduce `os.overhead`, patch driver bugs, or adjust thermal throttling curves on fielded devices.

Weakness W3 — Bypassed Industrial Process, Wafer Yield & Binning Contracts

- **Root Cause in Code:** `SPINOFF_SOURCE/js/phone-engineering.js` already defines rich deterministic contracts—`wafer()` (Murphy's yield model, gross/good/packaged dies, `aBin`/`bBin` split, scrap), `MATERIAL_GRADES`, `ROLE_CONTRACTS` (11 specialized roles: `fabless_soc`, `wafer_foundry`, `osat_packaging`, `dram_memory`, `nand_storage`, `camera_module`, `display_panel`, `battery_pack`, `pcb_rf`, `enclosure`, `os_platform`), and `evaluateRoleFeasibility()`—but `sim.js` largely bypasses them in favor of simplified single-stage batch counters.
- **Failure Mode:**
  - Foundries and OSATs lack finite equipment/cleanroom slot queues and process-specific defect-density learning curves $D_0(W_{\text{cum}})$.
  - Chip fabrication lacks die binning (`A-Bin` flagship dies vs. clock-binned `B-Bin` midrange dies) and product-line SKU tiering (`Base` vs. `Pro` storage/bin variants sharing one platform NRE).
  - Pre-silicon verification gaps do not distinguish metal-layer respins from full-mask respins.

Weakness W4 — Spot-Only Procurement Without the Pre-Launch Working-Capital Valley

- **Root Cause in Code:** Procurement settles spot purchases instantaneously against ready shelf stock without multi-week foundry/supplier capacity reservations, upfront escrow deposits, or net settlement obligations.
- **Failure Mode:** Hardware firms never experience the defining financial hazard of handset manufacturing—committing cash to wafer/component deposits 30–60 days before launch, suffering a `DVT`/`PVT` validation slip, and facing a real working-capital liquidity crunch while wages and supplier balances come due.

Weakness W5 — Single-Actor CEO Heuristics & Static Workplace Social Fabric

- **Root Cause in Code:** Strategic and project gating relies almost exclusively on `ceo = person(w, f.people[0])` personality scalars. Other employees act purely as hourly skill-point batteries (`60 * factor`).
- **Failure Mode:**
  - Missing C-suite multi-role tension: no explicit financial gatekeeper (`CFO` checking cash runway/escrow exposure), technical lead (`CTO` evaluating `Q_breakeven` and interface/thermal bottlenecks), or commercial lead debating trade-offs via persistent, inspectable `boardProposals`.
  - No intra-lab mentoring (senior specialists accelerating junior learning at a slight throughput cost) and weak coupling between chronic overwork/project cancellations and poaching susceptibility.

Weakness W6 — Omniscient Market Information, Missing Channels & Single-Timeline Observation

- **Root Cause in Code:** Consumers immediately know exact `consumerUtility(d, segment)` at the moment of release, all sales are direct-to-household single-unit transactions, and `Edit Hub` mutates live state in place without paired baseline comparison.
- **Failure Mode:**
  - No distinction between B2B channel `sell-in` (retailer/carrier batch orders with RF/network certification gates) and household `sell-through` (risking channel inventory overhang).
  - No information asymmetry: no supply-chain leaks triggering the Osborne effect (cannibalizing current-gen sales ahead of a major launch) and no independent teardown/review verification penalizing overhyped launch positioning.
  - Observers cannot easily inspect causal chains or compare a mutated experiment branch against an untouched control seed.

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2. Concrete Architectural Remedies & Engineering Specifications

Remedy R1 — Conservation-Exact Circular Macro-Cash Recycling (`v0.4.0`)

Maintain `totalCash(w) === w.openingCapital + w.editCapital` strictly via explicit `transfer()` calls—never mint satang:

1. **Personal Living Consumption Transfer:** During `daily(w)`, each person retains a prudence-weighted personal savings buffer `reserveSatang = baseReserve + profile.personality.prudence * prudenceScale`. A bounded daily fraction of cash above subsistence/meal reserves is transferred via `transfer(w, person, households, amount, 'รายจ่ายค่าครองชีพครัวเรือน')` and allocated across `w.segments` using seeded cohort income shares (using deterministic integer remainder distribution so $\sum \Delta \text{segment.cash} === \text{amount}$).
2. **Upstream Factor Income Recycling:** `w.treasury` represents upstream extraction, utilities, and external services whose workers are also consumers. Each day, a bounded velocity fraction (e.g., `1.5%–3.0%` of `treasury.cash` above an upstream liquidity floor) transfers via `transfer(w, w.treasury, w.households, recycleSatang, 'รายได้หมุนเวียนจากภาคชัพพลายเชนต้นน้ำ')` into `w.segments`.
3. **Updated Audit Equation:**
   - Replace static `s.cash + s.spent === s.initialCash` with: `s.initialCash + s.incomeReceived + s.refundsReceived - s.spent === s.cash`
   - Enforce `w.segments.reduce((acc, s) => acc + s.cash, 0) === w.households.cash`.

Remedy R2 — Cohort Vintage Installed Base, RMA Warranty Loop & OTA Patches (`v0.4.0`)

Keep time complexity $O(|S| \times |V|)$ by grouping active devices into weekly cohort vintage buckets rather than per-consumer objects:

1. **Data Schema (`w.installedBase`):**

```
JavaScript
{
  id: 'ib_104',
```

```
  segmentId: 'photo',
  designId: 'd12',
  firmId: 'rival-2',
  firmwareRev: 1,
  acquiredDay: 140,
  warrantyUntilDay: 505, // acquiredDay + 365
  activeUnits: 18,
  failedUnits: 0,
  retiredUnits: 0,
  healthPermille: 1000,  // integer 0..1000 for deterministic aging
  cumulativeThermalStress: 0
}
```

2. **Deterministic Aging & Hazard Model:**
   - Daily health decay scales with `1 / battery.cycleLife`, `max(0, evaluation.surfaceC - 41)`, and `1 / enclosure.dropResistanceScore`.
   - Daily field-failure count uses deterministic seeded binomial/Poisson-like integer draws against `activeUnits` and hazard rate h.
   - Units failing while `day(w) <= warrantyUntilDay` enter `firm.rmaQueue`. Units whose `healthPermille` drops below replacement threshold (or whose OS support expires past tolerance) transition to `retiredUnits` and add `replacementDemand` to that cohort's daily purchase intent.
3. **RMA Resolution & Root-Cause Learning:**
   - Resolving an RMA unit requires real paid `quality`/`manufacturing` work minutes plus either compatible replacement parts from `firm.stock` or a full/pro-rata satang warranty refund (`transfer(w, firm, w.households, refundSatang, ...)` credited back to `segment.cash` and `segment.refundsReceived`).
   - Unresolved RMAs erode `firm.brandTrust` (`0..100`), penalizing future `consumerUtility`.
   - Completed RMA teardowns generate subsystem-tagged `failureLessons` (e.g., `{ subsystem: 'thermal', defectCode: 'high_skin_temp', mitigationCredit: 12 }`) that reduce defect rates on future designs addressing that exact bottleneck.
4. **Post-Launch OTA Firmware Projects:**
   - Brands can schedule an `ota_patch` project consuming `software` hours to increment `design.firmwareRev`, updating active `installedBase` buckets to reduce `os.overhead`, fix `driverCoverage` deficits, or apply conservative power limits (`peakW` cap trading `taskScores.game` for lower `surfaceC` and lower RMA hazard).

Remedy R3 — Activate `phone-engineering.js` Role Contracts, Foundry Queues & Die Binning (`v0.5.0`)

1. **Multi-Stage Process Execution:** Connect `ROLE_CONTRACTS[role].processStages` (`workShare`, `passiveHoldMinutes`, `cleanroomRequired`) to supplier and in-house fab WIP batches.
2. **Foundry Capacity & D_0 Yield Learning:**
   - Foundry/OSAT firms expose finite line capacity (`waferStartsPerWeek`). External fabless/brand buyers and internal projects compete for queue slots via lead-time and price/deposit terms.
   - Track per-firm, per-node cumulative wafer volume W_{cum} to update `defectsPerCm2` in `E.wafer()`, bounded by a physical defect floor and subject to upward drift if maintenance expenses or `metrology` staffing lapse.
3. **Conserved Die Binning & Product Tiering (`Base` / `Pro`):**
   - `E.wafer()` outputs `{ aBin, bBin, scrap }`. Store `A-Bin` and `B-Bin` as distinct lot SKUs (`rev:A`, `rev:B` with reduced clock/power).
   - Enable brands to derive two commercial variants from one validated base platform (`Pro` using `A-Bin` + higher NAND/RAM SKU; `Base` using `B-Bin` + lower NAND/RAM SKU), unlocking realistic margin-mix strategy without duplicating full platform NRE.
4. **Tape-Out Verification & Respins:**
   - Track pre-tapeout verification coverage from `semiconductor`/`quality` hours. Insufficient coverage risks silicon errata during `validation`: minor errata require a lower-cost `metal_respin` (shorter delay), while severe architectural/interface errata require a `full_mask_respin` (`chip.maskCostSatang`).

Remedy R4 — Purchase Commitments, Escrow Ledger & Liquidity Management (`v0.5.0`)

1. **Escrow Account & Commitment Lifecycle:**
   - Add `w.escrow = { id: 'escrow', cash: 0 }` to `totalCash(w)` conservation (`treasury + households + escrow + sum(firms) + sum(people)`).
   - Large batch/foundry orders create a `purchaseCommitment`: buyer transfers `depositSatang` (e.g., `30%`) to `w.escrow` on booking; seller reserves capacity/inputs; upon lot completion, buyer transfers remaining `70%` to seller and `w.escrow` releases `depositSatang` to seller.
   - If the buyer defaults due to cash exhaustion during a launch delay, contract rules execute deterministically: `w.escrow` forfeits the penalty share to the seller, refunds any unspent remainder to the buyer, and releases unshipped stock for resale on the open market.

Remedy R5 — Multi-Role C-Suite Governance (`boardProposals`) & Lab Mentoring (`v0.5.0`)

1. **Dynamic Role Assignment:** Map staff in any firm (from 3-person SMEs to larger enterprises) to functional hats (`CEO`, `CFO`, `CTO`, `CommercialLead`) using existing `HumanCore` aptitudes, values, and subjects.
2. **Structured Board Proposals:** Major capital allocations (custom silicon vs. merchant buy, fab/line expansion, high-MOQ commitment, project cancellation) create an auditable `boardProposal`:
   - `CTO` scores technical differentiation, compatibility, and $Q_{\text{breakeven}} = \frac{F_{\text{NRE}}}{P_{\text{buy}} - V_{\text{make}}}$ (explicitly flagging non-positive denominators when $P_{\text{buy}} \le V_{\text{make}}$).
   - `CFO` enforces liquidity runway (`projectedMinCash >= wageRunwayDays * dailyWageBill + dueCommitments`, scaled by `prudence`).
   - `CommercialLead` evaluates cohort wallet depth, unsold inventory overhang, and competitor congestion.
   - Persist `votes`, `selectedAlternative`, `rejectedAlternatives`, and `dissentReasonTh` for direct inspection in the Thai UI.
3. **Mentoring & Morale Feedback:**
   - Co-located senior engineers (`skill >= 65`) boost junior learning rates (`skill <= senior - 15`) by `1.5x–2.2x` up to `profile.potential`, deducting `8%–12%` of the senior's direct task output as mentoring time.
   - High chronic stress, unpaid days, or repeated late-stage cancellations lower retention thresholds in `recruit()` without ever transferring employer-owned IP/designs with the departing person.

Remedy R6 — Channels, Certification Gates, Tech Leaks & Counterfactual Edit Hub (`v0.6.0`)

1. **B2B Channel / Carrier Layer (`Sell-In` vs. `Sell-Through`):**
   - Introduce seeded retail/carrier channel buyers with finite working capital and RF/reliability qualification thresholds (`pcb_rf.signalIntegrity`, `radioGeneration`, maximum failure hazard).
   - Channels buy wholesale lots (`sell-in` at wholesale discount), hold `channelInventory`, and sell through to household cohorts (`sell-through`). Channel stock overhang blocks reorders—never double-counting wholesale and retail sales.
2. **Imperfect Information, Leaks & Independent Reviews:**
   - Outsourcing EVT/DVT/PVT to external EMS/foundries carries a bounded leak probability. If an upcoming SKU with substantially higher utility leaks early, cohorts with high `openness`/performance preference defer current purchases (Osborne effect), slowing legacy SKU sell-through.

- Separate launch claims from post-release measured `mediaReviews` (evaluating sustained thermal performance, low-light camera, and real battery life), updating `firm.brandTrust`.

3. **Counterfactual Branching & Causal Watchlists in `Edit Hub`:**
   - Allow `Edit Hub` to trigger logged systemic shocks (e.g., temporary supplier cleanroom outage, raw-input supply contraction, cohort budget shift) and optional **Shadow Baseline Snapshots** (forking current world state before an `Edit Hub` intervention so the UI can display side-by-side deltas: *Control World* vs. *Experiment World* after N days).

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3. Phased Post-v0.3.0 Implementation Roadmap

Milestone	Schema / Key Target	Core Deliverables	Exit Gate & Soak Requirement
v0.4.0	`schema: 4` (`silicon-handset-v040`)	R1 Circular household/treasury cash flow; R2 Vintage `installedBase`, degradation, RMA warranty queue, teardown lessons, and post-launch OTA firmware patches; Thai lifecycle & cash-flow inspector UI	`5 seeds × 1,095 days (3 years)` soak with active sales in Year 3, exact satang/unit conservation, RMA & OTA causal tests
v0.5.0	`schema: 5` (`silicon-handset-v050`)	R3 `phone-engineering.js` multi-stage foundry/OSAT queues, D_0 yield learning, `A-Bin`/`B-Bin` lot splitting, `Base`/`Pro` SKU variants, respin classes; R4 `escrow` purchase commitments; R5	Multi-seed divergence proof (specialist supplier vs. merchant brand vs. custom-silicon firm), escrow default/liquidity crunch tests, zero double-labor

Milestone	Schema / Key Target	Core Deliverables	Exit Gate & Soak Requirement
		C-suite `boardProposals` & lab mentoring	
v0.6.0	`schema: 6` (`silicon-handset-v060`)	R6 Carrier/retailer `sell-in`/`sell-through` channels, RF certification gates, supply-chain leaks (Osborne effect), independent review calibration, causal watchlist & paired counterfactual diffing in `Edit Hub`	Full browser mobile (`390px` @ `150%` zoom) & desktop (`1280px`) accessibility/purity suite, counterfactual isolation tests, adversarial save tests

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4. Mandatory Verification & Delivery Protocol for Implementing Agents

Every patch (`v0.4.0+`) must execute and report real evidence under `SPINOFF_SOURCE`:

1. **Automated Verification Suite:**
   - `npm run check` — Syntax validation, initial world `audit()`, and `CORE_LOCK.json` SHA-256 verification of `human-core.js`.
   - `npm test` — Unit, causal counterfactual, conservation, and adversarial `deserialize()` tests (mid-RMA, mid-escrow, mid-foundry-queue, malformed/future fields rejected safely without clobbering active world or valid stored saves).
   - `npm run build` — Rebuild both standalone HTML aliases (`Silicon_Handset_single.html` and `SPINOFF_SOURCE/Silicon_Handset_single.html`) and verify byte equality.
   - `npm run test:browser` — Playwright Chromium verification at `390px` and `1280px` viewports, `150%` mobile zoom, dialog/tab keyboard accessibility, `Edit Hub` provenance display, and snapshot/hash read-purity under repeated UI tab switches.

    - `node tools/soak.cjs` — Multi-seed long-horizon soak (`>= 365` days, `1,095` days from `v0.4.0+`) with daily `audit(w)` and periodic `serialize`/`deserialize` round-trip equality checks.
2. **Documentation & Skill Synchronization:**
    - Update `spinoff_phone/PROJECT_STATUS.md`, `project_status_and_ai_onboarding.md`, `spinoff_phone/WORK_STATUS.md`, `spinoff_phone/IDEAS.md` (remove delivered items, retain unimplemented proposals), `.agents/skills/silicon-handset-dev/SKILL.md`, `.agents/skills/silicon-handset-every-patch/SKILL.md`, and `PATCH_REPORT_V0.X.0.md`.
    - Prepare updated Gemini app skill sync artifacts per `spinoff_phone/GEMINI_APP_SKILL_SYNC.md` and report sync status honestly.
3. **ZIP Handoff Integrity (When Delivering via Chat Archive):**
    - Single-root archive with explicit directory entries (including `.agents/`), excluding `.git`, `node_modules`, caches, and nested ZIPs; verify final archive via fresh extraction, CRC check, file hash manifest, and external SHA-256 receipt.